

Video Annotation System Using a Voice User Interface

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Abstract—In this study, a voice user interface (VUI) system for video annotation as a method that requires less time and offers high flexibility and accuracy is proposed. To evaluate the system performance, 87 information tags were recorded for basketball shooting scenes. The results demonstrate a tag error rate of 9.20% and a root mean square error of 1.36 s for input timing. The proposed system confirmed the feasibility of annotating videos using a VUI.

Index Terms—Sport video, data annotation, voice user interface, speech recognition

I. INTRODUCTION

In this study, a voice user interface (VUI) system for voice annotation of event timing and actions in sports games is proposed. Video analysis plays a crucial role in skill improvement in sports. The analysis primarily involves collecting statistical data on “when, who, and what” game actions to develop effective team strategies and training programs. Traditional manual analysis methods are labor intensive and often require specialized analysts for professional sports teams. To reduce the burden of annotation work, related research has proposed methods such as button input [1] and automatic video analysis systems [2]. However, these methods have limitations, such as limited input flexibility and recognition accuracy.

The proposed system provides a quick voice-based input method to add simple tags for “when, who, and what” to sports game videos. The flexibility of capturing “who and what” information is achieved by speech recognition, while “when” information is estimated from the start time of the input utterance. The proposed system reduces the annotation workload to the duration of the video and the time for correcting misrecognition, thereby significantly reducing the user’s workload.

II. SYSTEM OVERVIEW

To tag an event in a sports game video, the user speaks the uniform color and number of the player involved (player information) and information describing the event (scene information). The time synchronized with the video is obtained from the frame number at the start of the speech, and tags containing the time, player, and scene information are then added. Fig. 1 presents the overall architecture of the proposed method, and

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Fig. 2 illustrates the user interface. For implementation, the proposed system employs WebRTC voice activity detection (VAD) [3], for speech segment detection, faster-whisper [4] for speech recognition, chiVe [5] for word embedding, and a support vector machine for classification.

A. Correction of Misrecognition by Word Replacement

Preliminary experiments revealed that certain words were frequently misrecognized; thus, we implemented a function to replace those words with correctly recognized ones. By creating a word replacement list in advance, recognition accuracy can be improved.

III. EXPERIMENTS

We used the developed system to tag basketball shooting scenes and verify the accuracy of the time, player, and scene information.

A. Evaluation Metrics

We compared manually created ideal event data against the generated tags to calculate the word error rate (WER) and the tag error rate (TER). TER measures the percentage of tags with incorrect division segmentation between player and scene information. In addition, we calculated the distribution of the time deviation between the ideal event data and the generated tags, as well as the root mean square error (RMSE). The time deviation was calculated by subtracting the generated tag time from the ideal event time. The ideal event time was set to determine the shot result. The time deviation was caused by the reaction time when recognizing video events and determining the spoken content. Therefore, the extent to which the user’s reaction time affects the tag time information was also evaluated.

B. Data Acquisition

Three people with basketball experience tagged a 10-min game video containing 29 events, resulting in 87 total tagged instances.

C. Results

1) *WER and TER*: Table I presents the WER, TER, and the number of word substitutions, insertions, and deletions. After excluding the misrecognition caused by user omission and rephrasing tags during recording, the system achieved a WER

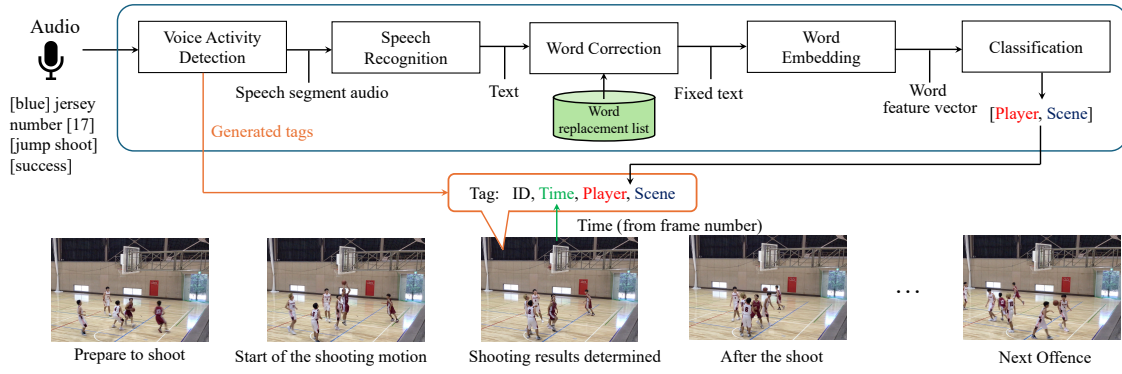


Fig. 1. System overview

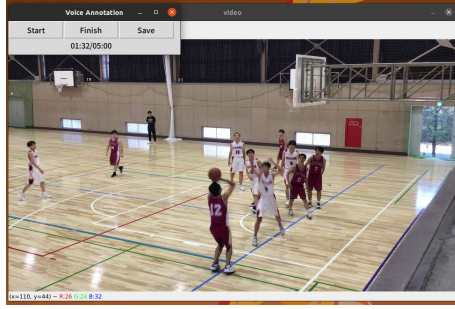


Fig. 2. Screenshot of our system

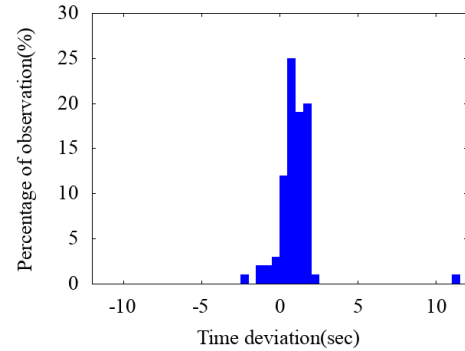


Fig. 3. Histogram of user response time

TABLE I
ACCURACY OF SPEECH RECOGNITION FOR THE GENERATED TAGS

WER(%)			TER(%)
12.0			9.20
Substitutions	Insertions	Deletions	
30	1	21	-

of 11.0% and a TER of 6.98%. To improve tag segmentation accuracy, the number of substitutions and deletions must be reduced. Word deletion occurs frequently at the end of the utterance.

2) *Time Deviation*: The RMSE of the timing accuracy was 1.36 s. Fig. 3 shows the time-deviation distributions, where the horizontal axis represents the time deviation (in seconds), and the vertical axis represents the percentage of observations. Negative time deviations occurred when the speech began while the shooting motion was in progress. A large deviation at the 10-s mark resulted from the user missing the target event and delaying speech. However, 98.8% of the tags had a time deviation within 3 s, the effect of individual user reaction time on the system is small.

IV. DISCUSSION

To improve tag segmentation accuracy, reducing word substitutions and deletions is essential. For substitutions, accuracy can be improved by expanding the word replacement list, particularly if typical misrecognition patterns are observed.

Word deletions occur when recognition fails at the speech recognition stage, making automatic accuracy improvement difficult and requiring manual correction. Word deletions typically occur at the end of utterances, indicating that VAD false

detection may have discarded the speech segments. In such cases, extending the allowable pause time may reduce word deletion.

V. CONCLUSION

In this study, a system that uses a VUI for video annotation to facilitate sports game video analysis was proposed. The experimental evaluation using a basketball game video demonstrated that the proposed system achieved a TER of 9.20% and an RMSE of 1.36 s. These results indicate that the proposed system is effective for VUI-based video annotations.

The proposed system is useful not only for basketball videos but also for videos of other sports (e.g., soccer) that use time information for event annotations.

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